

Seed and Event Sensitivity in Lightweight SAR Flood Change Detection: A Controlled Negative Result and an Accuracy–Cost Benchmark

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Abstract. Comparisons between deep models are often reported from a single training run. We study what that practice hides in one applied task: mapping floods from pairs of synthetic aperture radar (SAR) images with lightweight change detectors, where losses for small objects and boundary heads are natural additions. On the UrbanSARFloods dataset we train a 1.1M-parameter Siamese network in a 2×2 factorial (size-weighted loss, boundary package) together with two baselines, FC-Siam-Diff and a SAR adaptation of the Bitemporal Image Transformer, over six seeds and two training recipes, under rules fixed in advance, and evaluate all 36 runs once on a sealed test set. Under a 15-epoch regime without augmentation, we detect no reliable effect of either component: on test the paired contrasts are at most about 2 percentage points and of inconsistent sign, and every model recovers only 2–7% of small flooded regions. The ranking between model families is stable across seeds, whereas the ranking among the factorial variants is not. Seed-to-seed variation concentrates in a single event on each split, but a pre-specified explanation based on flood fraction is not supported by the test set. Parameter count, counted operations, GPU latency and CPU latency rank the models in four different orders. We close with reporting recommendations.

Keywords: SAR · flood mapping · change detection · lightweight models · seed sensitivity · negative result · efficiency

1 Introduction

A common way to evaluate a new loss or network component is to train one model with it and one without, and compare two numbers. It is well documented that such comparisons can be decided by the random seed [19, 3, 17] and that simple baselines often remain competitive once comparisons are controlled [7]. This paper is a worked case of that problem in an applied setting where it matters, because the evaluation data consist of only a few independent events.

The task is flood mapping from satellite radar. A synthetic aperture radar (SAR) sensor images the ground through cloud and at night, which makes it the usual data source while a flood is happening [24]. Given one image acquired before the flood and one during it, a model must output a mask of

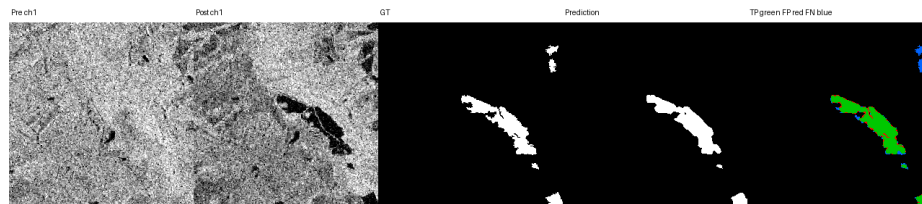


Fig. 1. The task. From left to right: radar backscatter (one polarization) before and during the flood, the flood label, a prediction, and the error map (green: correct flood; red: false alarm; blue: missed flood). A validation patch from the Lumberton event, variant A, shown to illustrate the task; it is one of the first patches by identifier and was not chosen to be representative. Note the small flooded regions that are missed.

flooded pixels; this is a *change-detection* problem (Fig. 1). Public datasets such as Sen1Floods11 [1] have made deep segmentation models the standard approach for open-area flooding. Urban flooding is harder: buildings produce strong radar returns that depend on geometry, flooded streets are narrow, and the flooded regions are often only a few pixels wide. The authors of UrbanSARFloods report that common training choices struggle with this imbalance and that urban flood detection remains challenging [24]; earlier work combined intensity with interferometric coherence to map urban floods [14].

Two further pressures shape model design for this task. First, flood maps are wanted quickly and, increasingly, close to the sensor, which favours small models [16, 6]. Second, the regions that matter most to responders are small, so it is tempting to add components that target them: losses that up-weight small connected components [13, 18] and heads or losses that sharpen boundaries [12].

This paper started from exactly that design: a 1.1M-parameter Siamese network with a size-weighted Tversky loss and a thin boundary head. A single-seed pilot gave results that could be read as a trade-off between the two components. When we repeated the comparison under protocols fixed in advance, with new seeds, a second training recipe and finally a sealed test set, the component effects did not replicate. We therefore report the study as it is: a controlled comparison in which no reliable benefit was detected, together with what it shows about seed and event sensitivity and about the cost of the models involved.

We ask three questions. **RQ1:** do size weighting and a boundary package have a consistent effect across seeds, two training recipes and a held-out test set? **RQ2:** how stable are method rankings against seed and event variation? **RQ3:** what is the accuracy–cost trade-off of sub-5M-parameter change detectors on a GPU and on a CPU?

Our contributions are findings rather than a new method:

- A controlled negative result for both components across six seeds, two recipes and one sealed test evaluation: no reliable effect was detected, and any effect on test is at most about two percentage points (Sect. 4.1).

- An analysis of seed and event sensitivity, including a pre-specified explanation that the test set did not support (Sects. 4.2–4.4).
- An accuracy–cost benchmark of three model families with several cost indicators measured on two devices (Sect. 4.5).

All conclusions are conditional on the training regime we used (15 epochs, no data augmentation, training from scratch) and on one dataset. We state these limits where each claim is made.

2 Related Work

SAR flood datasets. Sen1Floods11 provides georeferenced Sentinel-1 data for 11 flood events [1]. Kuro Siwo covers 43 events and provides both ground-range and single-look complex products [2]. UrbanSARFloods contains 8,879 chips of 512×512 pixels from 18 events, with pre- and co-event Sentinel-1 intensity and coherence and separate labels for flooded open areas and flooded urban areas [24]. We use its intensity channels only.

Bitemporal change detection. Flood mapping from a pre-event and a co-event image is a change-detection problem. Fully convolutional Siamese networks [8] and transformer-based models such as the Bitemporal Image Transformer (BIT) [5] are common starting points, and Siamese and transformer-based designs have been applied to SAR flood pairs [21, 20]. Two recent studies caution that architectural novelty explains less than it appears to: a plain U-Net remains a top performer on standard optical benchmarks [7], and a broad benchmark finds Siamese U-Nets competitive once efficiency is counted [23].

Small regions and boundaries. The Tversky loss trades false positives against false negatives [22]. Instance-aware losses target the detection of individual, often small, instances that pixel-level losses can overlook [13, 18], and boundary losses address highly unbalanced segmentation [12]. These ideas were developed mainly for medical images; small changed regions have also been studied in optical remote sensing [4]. Validation guidelines recommend choosing metrics from the problem at hand, including object-level validation where individual objects are of interest [15].

Efficiency. Lightweight change detectors report strong accuracy with few parameters [6], including for SAR flood mapping [11], and flood segmentation from optical imagery has been demonstrated for on-board processing on a small satellite [16]. Dehghani et al. show that cost indicators such as parameters, FLOPs and speed can disagree and argue for reporting several of them [9].

Variance across runs. Reimers and Gurevych show that the random seed alone can change conclusions for sequence taggers [19]; Bouthillier et al. quantify several sources of variance in benchmarks [3]; and Picard documents seed effects in computer vision [17].

Table 1. Event-held-out split. “Positive” is the share of valid pixels labelled as flood. The training positive share was not recorded in our result records.

Split	Events	Patches	Positive	Components	Interior	small
Train	12	19,166	–	52,516	–	–
Validation	3	4,767	1.08%	6,745	370	370
Test	3	9,204	1.22%	27,175	4,899	4,899

Gap. Our literature search was bounded (one search engine, 31 queries on 3 October 2026 and follow-up queries on 10 October 2026) and mostly read at abstract level. Within those searches we found no SAR flood study that reports recall for small flooded regions inside a patch, or that reports how method rankings change across training seeds. This is a statement about our searches, not a claim that no such study exists.

3 Experimental Design

3.1 Data and Splits

We use the Sentinel-1 intensity channels of UrbanSARFloods [24]: backscatter in decibels in two polarizations (VH and VV) for the pre-event and co-event dates. The dataset’s two flood classes, flooded open areas and flooded urban areas, are merged into one binary flood class. We therefore do not measure urban-specific performance; “small” below refers to the size of a flooded region, not to its land-cover class. Scenes are cut into non-overlapping 256×256 patches, and each input channel is standardized with statistics fitted on training pixels.

Events, not patches, are held out (Table 1). The dataset’s test events stay as test; the remaining events are re-partitioned into 12 training and 3 validation events by a fixed hash rule. *This split, the binary target and the patch size all differ from the setup of the dataset paper, so our numbers cannot be compared with its numbers.* The dataset paper’s qualitative finding, that urban and small flooded areas are the hard part, is consistent with what we observe. A connected component is *small* if its area within the patch is at most 35 pixels, the 25th percentile of training components, and *interior* if it does not touch the patch border or an invalid pixel.

The validation events are Lumberton (2016), Coraki (2022) and Hebei (2023). The test events are Weihui (2021; 1.03% positive), Nova Kakhovka (2023; 1.86%) and Jubba (2023; 0.85%). Nova Kakhovka contains 4,379 of the 4,899 interior small components of the test set.

3.2 Models

SmallFlood-CDNet (1.08M parameters) is a Siamese encoder–decoder. A shared MobileNetV3-small encoder [10], trained from scratch with a two-channel input,

extracts features at strides 4, 8, 16 and 32 from each image. At every scale the absolute difference and the element-wise product of the two feature maps are concatenated and projected to 24, 32, 64 and 96 channels. A decoder with three upsample-and-skip stages built from depthwise-separable convolutions produces a stride-4 feature map, from which a convolutional head predicts change logits that are upsampled bilinearly to the input size. The loss is $0.4 \text{ BCE} + 0.4 \text{ Tversky}$, where BCE is binary cross-entropy and the Tversky loss [22] uses false-positive and false-negative weights of 0.3 and 0.7. We study two optional components in a 2×2 factorial:

- **Size weighting.** Positive pixels of a flooded region with area a receive weight $1 + \min(4, (65/a)^{0.5})$ in both loss terms, where 65 pixels is the median area of training regions.
- **Boundary package.** A boundary head (one depthwise-separable convolution and a 1×1 convolution) predicts a boundary map from the decoder features. A second 1×1 convolution maps the decoder features and the boundary probability to a residual, which is scaled by a learnable scalar (initially 0.1, kept in $[0, 1]$) and added to the change logits. The boundary map is supervised with a binary cross-entropy plus Dice loss of weight 0.2 on the one-pixel morphological gradient of the label.

Variant A uses neither, B size weighting, C the boundary package and D both.

The baselines are *FC-Siam-Diff* [8] with encoder widths 16, 32, 64 and 128 (0.49M parameters) and *BIT-SAR v2* (3.49M parameters). The latter is the Bitemporal Image Transformer (BIT) [5] in its published configuration with eight decoder blocks, adapted to SAR: the input stem takes two channels, unused classification layers are removed, all weights are initialized randomly, and the two-class output is converted to one change logit. Both baselines use $0.5 \text{ BCE} + 0.5 \text{ Tversky}$. The comparison between families is a *whole-method* comparison: besides architecture, the baselines differ from SmallFlood-CDNet in loss coefficients, initialization scheme and training script, and the three families did not receive the same tuning effort (Sect. 3.3).

3.3 Training

All runs use AdamW with weight decay 10^{-4} , gradient clipping at 1.0, batch size 8, 15 epochs, 32-bit precision and no data augmentation. Training is fully deterministic: the same seed reproduces the same weights. Within a seed, variants A–D start from the same initial weights (apart from the boundary head) and see the same batch order, so differences between them are paired.

We use two recipes. In the *original* recipe, SmallFlood-CDNet and FC-Siam-Diff use a learning rate of 3×10^{-4} and BIT-SAR v2 uses 10^{-4} . In recipe *R1* all models use 10^{-4} . R1 was selected for variant A in a separate screening study that compared two learning rates and two schedules on an internal development split of training events only (the 2019 Canada and Japan events), without loading the validation events. It was not tuned for the other variants or for the baselines,

and the learning rate of BIT-SAR v2 was fixed at 10^{-4} from the start. The study comprises 36 full training runs (six models, six seeds) plus 12 screening runs, on one NVIDIA RTX A6000.

This regime is short and uses no augmentation. Final training loss was nearly identical across seeds within each model (for example 0.167–0.172 for variant A under the original recipe), but we did not study longer schedules, and our conclusions hold for this regime.

3.4 Metrics

The primary metric is *event-macro F1*: pixel counts are pooled within each event and the per-event F1 scores are averaged. We also report pooled pixel F1; component F1 with one-to-one matching at intersection-over-union (IoU) ≥ 0.1 ; recall of small components, interior and clipped; boundary F1 with a two-pixel tolerance; and the IoU of a two-pixel inner boundary band. The decision threshold is fixed at 0.5, and we report no threshold-free measure on validation or test; Sect. 4.3 returns to this. Results are patch-level; scenes are not reassembled. With three seeds per recipe we summarize spread by the range (maximum minus minimum) and attempt no variance estimate.

The primary checkpoint is the last one (epoch 15). Selecting the best validation epoch chose transient early peaks in several runs, so best-epoch results are kept only as secondary information.

3.5 Staged Protocol and Decision Rules

The study proceeded in stages (Table 2). The discovery stage was exploratory. From the confirmation stage onward, each stage had a written protocol fixed before any run of that stage. Stage outcomes were judged by sign-based rules over seeds:

- *Size weighting does not improve interior-small recall*: supported if $B - A \leq 0$ and $D - C \leq 0$ in every seed, contradicted if both are positive in every seed, otherwise inconsistent.
- *The boundary package gives no inner-band gain*: the same with $C - A$ and $D - B$.
- *FC-Siam-Diff is not worse than the heavier models*: supported if FC-Siam-Diff is at least as good as A and as BIT-SAR v2 in event-macro F1 in every seed, contradicted if it is worse than the same comparator in every seed, otherwise inconsistent.

No statistical test is applied to pixels, patches or components, because they are not independent replicates. These rules are strict by construction: “supported” and “contradicted” each require six paired contrasts of one sign, so noise alone makes “inconsistent” a likely outcome. We therefore also report the individual contrasts (Table 4).

Three disclosures apply. First, the sign rules were written after the discovery seed had been seen and were applied only to later seeds. For the original-recipe

Table 2. Stages of the study. Each stage used seeds with no earlier results.

Stage	Recipe	Seeds	Evaluated on
Discovery	original	42	validation
Confirmation	original	1337, 2026	validation
Recipe screening (variant A)	four candidates	101–103	internal development split
Recipe confirmation	R1	201–203	validation
Sealed test, all 36 runs	both	all six	test, once

confirmation they used the best checkpoint for two of the three rules and only the two confirmation seeds; all later stages use the last checkpoint and three seeds. Second, the test evaluation was specified in full, including the list of 72 checkpoints (last and best of 36 runs), before any test array was opened, and it was run once. Third, input normalization statistics were fitted on all training events, including the two development events used for recipe screening.

3.6 Cost Measurement

Following Dehghani et al. [9] we report several indicators for one pre/post pair of 256×256 patches: parameters; floating-point operations (FLOPs) counted for convolutions and matrix multiplications only; serialized weight size; peak GPU memory; batch-1 latency in PyTorch on an NVIDIA RTX A6000 at 32-bit (FP32) and 16-bit (FP16) precision (100 warm-up and 500 timed runs, three repetitions); and batch-1 latency of the model exported to the Open Neural Network Exchange (ONNX) format, run in ONNX Runtime on an Intel Xeon Silver 4310 server processor with one and four threads (20 warm-up and 100 timed runs, three repetitions). The server was shared. GPU repetitions ran only when no other process used the GPU, apart from an idle process holding 314 MB at 0% utilization during our measurements. Both devices are server hardware. We did not measure an edge device, and we make no claim about one.

4 Results

4.1 The Two Components Have No Consistent Effect

Table 3 gives the outcome of the three rules on validation and on test for both recipes. Both component rules are inconsistent in all four settings.

“Inconsistent” must not be read as “no effect”. Table 4 lists the paired test contrasts. For size weighting, 8 of the 12 contrasts are positive, with a mean of +0.7 pp and a largest value of +2.3 pp; one contrast of -0.08 pp is what makes the R1 outcome inconsistent. For the boundary package, 9 of 12 are positive, with a mean of +0.4 pp. Our reading is therefore narrow: we detected no reliable effect, and if an effect exists on test it is at most about two points. This is far from the benefit that would justify the components, but it is not evidence that they do nothing.

Table 3. Rule outcomes and the range of per-seed contrasts in percentage points. Size: interior-small recall, $B - A$ and $D - C$. Boundary: inner-band IoU, $C - A$ and $D - B$. Baseline: event-macro F1, FC-Siam-Diff (FC) minus A and minus BIT-SAR v2 (BIT). Validation with the original recipe uses two confirmation seeds, and the best checkpoint for the size and baseline rules (Sect. 3.5); all other entries use the last checkpoint.

Setting	Size weighting	Boundary package	FC not worse (vs. A; vs. BIT)
Validation, original	inconsistent [−10.0, +7.0]	inconsistent [−3.2, +2.1]	inconsistent [+2.7, +5.5]; [−5.6, +2.8]
Validation, R1	inconsistent [−1.1, +10.5]	inconsistent [−0.4, +6.7]	inconsistent [−3.0, +24.4]; [−27.8, +2.0]
Test, original	inconsistent [−0.4, +2.3]	inconsistent [−1.1, +2.2]	contradicted [+6.5, +13.0]; [−4.0, −2.2]
Test, R1	inconsistent [−0.1, +1.8]	inconsistent [−0.4, +0.9]	contradicted [+1.0, +14.1]; [−14.3, −3.8]

Table 4. Paired test contrasts per seed in percentage points, last checkpoint. Size weighting: interior-small recall. Boundary package: inner-band IoU.

Recipe	Seed	Size weighting		Boundary package	
		$B - A$	$D - C$	$C - A$	$D - B$
Original	42	−0.35	+0.12	+0.41	+0.03
	1337	−0.12	−0.12	+1.62	+2.19
	2026	+1.20	+2.31	−1.09	+0.17
R1	201	+1.71	+0.88	−0.33	+0.28
	202	+0.43	−0.08	+0.28	+0.54
	203	+1.18	+1.76	−0.42	+0.91
Positive contrasts		8 of 12		9 of 12	
Mean		+0.74		+0.38	

The baseline rule is contradicted on test because BIT-SAR v2 exceeds FC-Siam-Diff in every seed of both recipes. FC-Siam-Diff in turn exceeds variant A in all six seeds.

The clearest evidence is in Table 5: with 4,899 interior small regions in the test set, every model recalls between 2.4% and 6.7% of them on average, with or without size weighting. Two cautions apply. Nova Kakhovka holds 89% of those regions, so the test evidence on small regions rests mainly on one event. And a recall this low may itself be a product of the training regime; a loss re-weighting has little to act on when the model finds almost none of the regions.

4.2 Seed Sensitivity and Rank Stability

Figure 2 shows event-macro F1 for every run. On validation, the spread within one model and recipe reaches 26.7 pp (FC-Siam-Diff under R1: 62.3, 83.4 and 56.7) and 16.0 pp for variant C. The size-weighting contrast $B - A$ on interior-small recall, measured at the best checkpoint, was −13.2 pp in the discovery seed

Table 5. Test results at the last checkpoint: mean over three seeds, with the range (max – min) in brackets, in percent. “Int. small” is interior-small recall and “Band IoU” the inner-band IoU.

Recipe	Model	Event F1	Pixel F1	Comp. F1	Int. small	Bound. F1	Band IoU
Original	A	62.0 (4.1)	49.9 (7.8)	19.3 (1.9)	3.0 (1.0)	40.1 (2.3)	15.5 (1.1)
	B	55.3 (10.3)	46.3 (6.3)	19.1 (1.0)	3.2 (1.1)	37.4 (1.2)	14.3 (0.3)
	C	62.2 (10.0)	55.6 (13.9)	18.9 (4.8)	2.7 (1.8)	40.0 (9.8)	15.8 (3.3)
	D	61.3 (10.5)	52.8 (13.6)	19.2 (1.9)	3.4 (1.6)	39.2 (6.7)	15.1 (2.5)
	FC-Siam-Diff	71.0 (4.5)	61.1 (15.3)	19.2 (4.1)	6.7 (0.7)	50.2 (5.8)	26.3 (4.5)
	BIT-SAR v2	73.9 (4.1)	70.0 (4.2)	27.1 (4.9)	5.1 (2.9)	52.1 (6.0)	19.8 (1.2)
R1	A	60.0 (8.6)	45.6 (6.0)	17.9 (2.7)	2.4 (1.1)	37.9 (7.2)	14.3 (3.1)
	B	58.3 (6.2)	44.1 (6.8)	18.9 (0.9)	3.5 (0.2)	38.1 (2.3)	13.9 (0.9)
	C	59.8 (11.9)	47.7 (11.0)	18.0 (4.3)	2.7 (2.8)	37.6 (10.1)	14.1 (3.7)
	D	59.4 (4.9)	48.1 (3.8)	19.1 (1.9)	3.5 (1.8)	38.7 (2.9)	14.5 (1.2)
	FC-Siam-Diff	65.5 (6.4)	59.5 (11.8)	15.9 (4.4)	4.7 (4.4)	44.7 (7.6)	23.1 (4.4)
	BIT-SAR v2	74.2 (8.5)	70.5 (9.2)	25.6 (10.2)	4.0 (4.1)	51.4 (13.6)	19.5 (3.7)

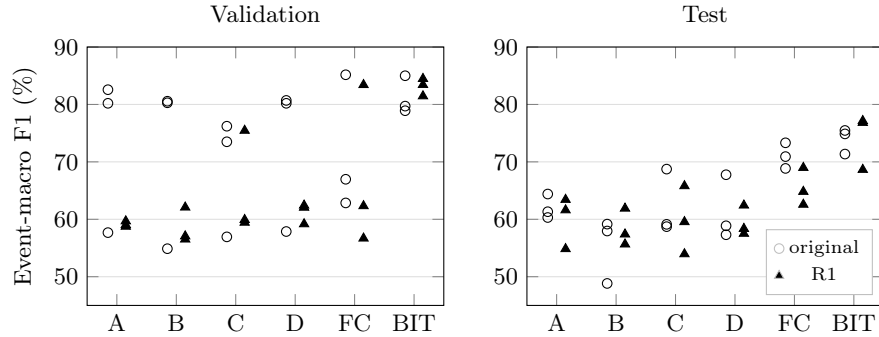


Fig. 2. Event-macro F1 of every run at the last checkpoint. Each marker is one seed (circles: original recipe; triangles: R1). FC: FC-Siam-Diff; BIT: BIT-SAR v2.

and +1.6 and +7.0 pp in the two confirmation seeds: a different sign in the seed that prompted the hypothesis than in the seeds that tested it.

On test we counted, for each of the 15 pairs of models, whether their order by event-macro F1 is the same in all three seeds of a recipe. Ten pairs are stable under the original recipe and nine under R1. The pattern is clear. The order *between* families is stable: BIT-SAR v2 is above FC-Siam-Diff in all six seeds, and both baselines are above each SmallFlood-CDNet variant with one exception (variant C exceeds FC-Siam-Diff in one R1 seed, 65.8 against 64.8). The order *among* variants A–D is stable for only one pair under each recipe. A study that ranked A–D from one seed would have reported an ordering that the next seed reverses.

The family order should be read with the caveat of Sect. 3.3: it compares whole methods that differ in more than architecture and that were not tuned equally.

4.3 Variation Concentrates in One Event, but Flood Fraction Does Not Explain It

On validation, almost all of the seed-to-seed spread comes from one event. Coraki and Hebei are comparatively stable for every model (F1 of about 90–95% and 68–86% under R1), whereas Lumberton F1 ranges from below 10% to above 80%. Under R1, variant A is below 20% F1 on Lumberton in 42 of its 45 training epochs across the three seeds. The failure is over-prediction: in epochs 11–15 the SmallFlood-CDNet variants predict 1.5–4.4% of pixels as flood where the truth is 1.08%, while training loss is almost identical across seeds. BIT-SAR v2 never falls below 20% on Lumberton in any of its 90 epochs across the six seeds.

Lumberton has a low flood fraction (about 0.44% of valid pixels, against 1.08% for the validation set as a whole), so we wrote down a rule before opening the test set: the finding *generalizes* if the test event with the lowest flood fraction has the largest across-seed F1 range in at least 7 of the 12 model-and-recipe groups, and *does not generalize* if in at most 4. The low-fraction test event is Jubba, and it has the largest range in 4 groups. The outcome is “does not generalize”.

Variation on test is still concentrated in one event, but it is Nova Kakhovka, the event with the *highest* flood fraction and almost all of the small components. It has the largest range in the other 8 groups. All 14 runs with a test event below 20% F1 have Nova Kakhovka as that event, all 14 are SmallFlood-CDNet variants, and in 12 of them the model predicts more flood pixels than the truth. No FC-Siam-Diff or BIT-SAR v2 run falls below 20% on any test event. We can therefore say that, at a fixed threshold of 0.5, seed sensitivity took the form of over-prediction on one event per split for the smallest model family. We cannot say which property of an event causes it; flood fraction is not it. Nor can we say whether the affected runs rank pixels poorly or are only miscalibrated on that event, because we report no threshold-free measure on validation or test. Miscalibration under event shift, under-training, the false-negative weight of the Tversky loss and the lack of augmentation all remain untested explanations.

4.4 A Recipe Chosen on Held-Out Training Events Did Not Transfer

Because the original recipe collapsed on Lumberton in one of three seeds, we screened four recipes for variant A on two training events held out as a development split. The lower learning rate with a constant schedule (R1) gave the highest and most balanced development score in all three seeds (mean event-macro F1 50.2% against 37.4% for the original recipe) and met a stability rule fixed in advance.

On the real validation events R1 did not help. Variant A reached 58.7, 59.0 and 59.7% event-macro F1 in its three R1 seeds, against 82.6, 80.2 and 57.7% for three other seeds under the original recipe, with Lumberton F1 of 8.8, 3.8 and 6.1%. The comparison is not paired, since the seeds differ, but the direction is plain: the lower learning rate made the failure consistent instead of removing it.

Table 6. Cost for one pair of 256×256 patches. FLOPs count convolutions and matrix multiplications only. GPU: PyTorch on an RTX A6000, batch 1, median latency at 32-bit (FP32) and 16-bit (FP16) precision. CPU: ONNX Runtime on a Xeon Silver 4310, batch 1, median latency with one and four threads. Size and peak memory are for 32-bit weights.

Model	Params	FLOPs	Size	GPU (ms)		CPU (ms)		Mem.
	(M)	(G)	(MB)	FP32	FP16	1 thr.	4 thr.	(MB)
SmallFlood-CDNet (A, B)	1.08	0.35	4.5	8.7	10.5	10.7	7.5	10.0
+ boundary head (C, D)	1.08	0.35	4.5	9.2	10.8	10.6	8.6	10.0
FC-Siam-Diff	0.49	7.02	2.0	3.0	3.8	81.4	26.0	56.0
BIT-SAR v2	3.49	21.43	14.1	24.5	25.7	350.9	163.3	83.4

On test the two recipes are close for variant A (62.0% and 60.0%), R1 is worse for FC-Siam-Diff (71.0% and 65.5%), and BIT-SAR v2, which used the same learning rate in both, is unchanged (73.9% and 74.2%).

This stage also exposed a weakness in our own stability rule, which compared each seed with the median over seeds. A model that fails in every seed passes such a rule, and variant A under R1 was labelled “stable” at a Lumberton F1 below 10%. We kept the rule as written and report what the label means here: consistent, not good. A stability rule needs an absolute floor or a reference model.

4.5 Accuracy and Cost

Table 6 shows that the four usual indicators rank the models differently. By parameters, FC-Siam-Diff is the smallest. By counted FLOPs, SmallFlood-CDNet is lightest by a factor of 20 over FC-Siam-Diff and 62 over BIT-SAR v2. By GPU latency, FC-Siam-Diff is fastest and SmallFlood-CDNet is 2.9 times slower despite its low FLOP count. By CPU latency, SmallFlood-CDNet is fastest: on one thread FC-Siam-Diff is 7.6 times and BIT-SAR v2 32.8 times slower. This is the pattern that Dehghani et al. [9] warn about, observed here within a single task. We did not investigate why the model with the fewest counted operations is slower on the GPU, and we offer no explanation.

Half precision did not reduce batch-1 latency on this GPU for any model, although it roughly halved the stored size and reduced peak memory. At batch size 8 it raised throughput for FC-Siam-Diff (1,020 to 1,662 pairs per second) and BIT-SAR v2 (233 to 291) but not for SmallFlood-CDNet (708 to 566). On the validation split, 16-bit inference agreed with 32-bit inference on at least 99.989% of valid pixels and changed no reviewed metric by more than 0.18 pp; ONNX Runtime agreed with PyTorch on all pixels of the 64 patches we compared. Timing on a shared server is noisy: the three repetition medians were 8.73, 8.70 and 8.67 ms for SmallFlood-CDNet at FP32, but 3.90, 2.59 and 3.84 ms for FC-Siam-Diff at FP16. We report the values as measured and did not repeat measurements.

Figure 3 combines Tables 5 and 6. The model that is cheapest on a CPU is the least accurate, and the model that is most accurate on test, and the only

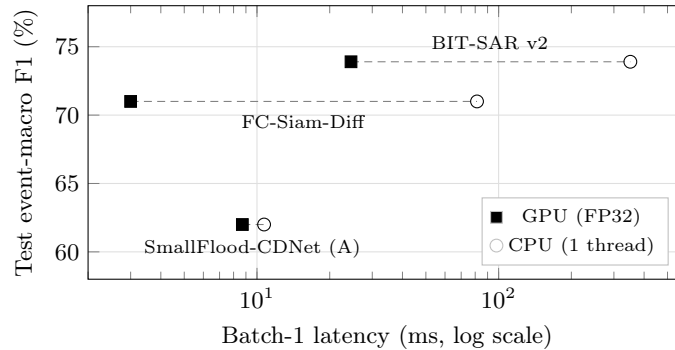


Fig. 3. Accuracy against latency. Test event-macro F1 is the mean over three seeds under the original recipe (Table 5); latency is from Table 6. Each model appears twice, once per device, joined by a dashed line.

one that never collapsed on a validation event, is the most expensive on every indicator. FC-Siam-Diff sits between them, with the caveat that its accuracy depends more on the recipe. The result of this section is that cost indicators disagree, not that a particular model suits edge deployment: both devices are server hardware, and the accuracy ranking is a whole-method comparison.

5 Discussion

What the negative result means. The component effects that looked interpretable in one seed are within the variation between seeds. The discovery seed suggested that size weighting *reduced* interior-small recall by 13 points; later seeds showed changes of either sign, and on a test set with thirteen times more small components the contrasts are within about two points, most of them slightly positive. We do not conclude that size weighting or boundary supervision cannot help. We conclude that, for this model, dataset and 15-epoch regime, no reliable benefit could be shown, that any benefit on test is small, and that a single-seed study would have reported one effect or its opposite depending on the seed. A longer or augmented regime in which the models find more of the small regions could give a different answer.

What is not established. We do not know why one event per split triggers over-prediction in the smallest model family. We tested and rejected one explanation, flood fraction; calibration at a fixed threshold is a second one that our metrics cannot test. We also do not know why BIT-SAR v2 is robust. It differs from the other models in architecture, initialization and loss coefficients at once, so its advantage cannot be attributed to any one of them. Separating these factors is a natural next study.

Recommendations. For studies of this kind we suggest: (i) report several seeds with the range, not one run; (ii) report per-event results, because an event-macro score over three events can be decided by one of them; (iii) report the last checkpoint next to the best one; (iv) report object-level metrics with their denominators, since pixel F1 hides a 2–7% small-region recall; (v) report several cost indicators on more than one device; and (vi) write the comparison rule down before looking, and say when it was written.

Limitations. The study uses one dataset, three validation events and three test events, and three seeds per recipe; the sign rules are conventions, not statistical tests, and they are strict enough that small consistent effects would be labelled inconsistent. Test evidence on small regions comes mainly from one event. The training regime is short and unaugmented, the decision threshold is fixed, and the model families were not tuned equally. Inputs are intensity only, although coherence has been used for urban floods [14]. Evaluation is patch-level, the two flood classes are merged, and the split is our own, so no comparison with published numbers is possible. The sign rules were written after a discovery seed, and one stability rule had the loophole described in Sect. 4.4. Cost was measured on a shared server and not on edge hardware, and the FLOP counter ignores normalization, activation and attention-softmax operations. Our literature search was bounded.

6 Conclusion

We set out to improve the detection of small urban floods in SAR imagery with a lightweight model, and instead documented three things. First, under a 15-epoch regime, neither a size-weighted loss nor a boundary package showed a reliable effect across six seeds, two recipes and a sealed test set; any effect on test was at most about two points, and all models recovered only a few percent of small flooded regions. Second, rankings between model families, compared as whole methods, were stable while rankings among close variants were not, with the variation concentrated in one event per split for reasons that flood fraction does not explain. Third, parameters, FLOPs, GPU latency and CPU latency ordered the models differently, so “lightweight” needs a device and an indicator attached to it.

The open questions are specific: what makes the BIT-based model robust, whether the findings hold on a second SAR flood dataset, whether coherence changes the picture for urban floods, and what the models cost on an actual edge device.

Data and code availability. UrbanSARFloods is publicly available from its authors [24]. The training and evaluation code, the written protocols and the per-run result records of this study are available from the corresponding author on reasonable request.

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